

FIELD AND GALOIS THEORY

Anita Naolekar, notes by Ramdas Singh

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Chapter 1

ON FIELDS

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Quizzes will account for 20% of the final grade. The mid-sem exam will account for 30% of the final grade, and the final exam will account for 50%.

1.1 An Introduction

A *field* is a set F with two binary operations, addition and multiplication, satisfying the following axioms:

1. $(F, +)$ is an abelian group with (additive) identity element 0.
2. $(F \setminus \{0\}, \cdot)$ is an abelian group with (multiplicative) identity element 1.
3. Multiplication is distributive over addition, i.e., for all $a, b, c \in F$, we have $a \cdot (b + c) = a \cdot b + a \cdot c$.

For a positive integer n , define $n \cdot 1 = \underbrace{1 + 1 + \cdots + 1}_{n \text{ times}}$. The *characteristic* of a field F is defined as the smallest positive integer p such that $p \cdot 1 = 0$, if such an p exists; otherwise, the characteristic is defined to be 0.

Proposition 1.1. *The characteristic of a field is either 0 or a prime number.*

Proof. Let F be a field with characteristic $\text{char } F = n > 0$. Let $n = ab$ with $1 < a, b < n$. Then $0 = n \cdot 1 = (ab) \cdot 1 = (a \cdot 1)(b \cdot 1)$. Since F is a field, it has no zero divisors, so either $a \cdot 1 = 0$ or $b \cdot 1 = 0$. This contradicts the minimality of n , so n must be prime. ■

Definition 1.2. Let F be a field. The *prime subfield* of F is the smallest subfield of F which contains the multiplicative identity 1.

This prime subfield can be viewed as both the intersection of all subfields of F , and the subfield generated by 1.

Define, for a positive integer n , $(-n) \cdot 1 = -(n \cdot 1)$. Now define $\varphi : \mathbb{Z} \rightarrow F$ by $n \mapsto n \cdot 1$ for all $n \in \mathbb{Z}$. It is easy to see that φ is a ring homomorphism. It is also routine to check that if F has characteristic $p > 0$, then $\ker \varphi = p\mathbb{Z}$, and this induces an injective ring homomorphism $\tilde{\varphi} : \mathbb{Z}/p\mathbb{Z} \rightarrow F$. If F has characteristic 0, then $\ker \varphi = \{0\}$, and $\mathbb{Z} \hookrightarrow F$ via φ . Since F is a field, it must contain a copy of $\mathbb{Q}$ as well. In the earlier case where F has characteristic $p > 0$, then we obtained $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z} \hookrightarrow F$. Hence, for a field of characteristic 0, the prime subfield is isomorphic to $\mathbb{Q}$, and for a field of characteristic $p > 0$, the prime subfield is isomorphic to $\mathbb{F}_p$.

1.2 Extensions

Let F be a field. A *field extension* of F is a field K such that $F \subseteq K$. We denote this by K/F . Note that K may also be viewed as a vector space over F . The *degree* of the extension K/F is defined as the dimension of K as a vector space over F , and is denoted by $[K : F]$. If $[K : F]$ is finite, we say that K/F is a *finite extension*. Otherwise, we say that K/F is an *infinite extension*.

The following is the crux of the theory of field extensions; given any polynomial $f(x) \in F[x]$, does there (always) exist an extension of F in which $f(x)$ has a root? Note that it is enough to consider irreducible polynomials, since if $f(x) = g(x)h(x)$, then any root of $g(x)$ or $h(x)$ is a root of $f(x)$. The following theorem answers this question in the affirmative.

Theorem 1.3. *Let $f(x) \in F[x]$ be a non-constant irreducible polynomial. Then there exists a field extension K/F such that $f(x)$ has a root in K .*

Proof. Let $K = F[x]/(f(x))$. Since $f(x)$ is irreducible, $(f(x))$ is a maximal ideal of $F[x]$, and hence K is a field. Look at the composition

$$F \hookrightarrow F[x] \twoheadrightarrow F[x]/(f(x)) = K. \quad (1.1)$$

This complete composition, φ , is a field homomorphism, so the kernel must be trivial or the homomorphism is injective. Since $\varphi(1) = 1 + (f(x)) \neq 0 + (f(x))$, we see that $\ker \varphi = \{0\}$, and hence F is embedded in K ; K/F is a valid field extension. We claim that $\alpha = x + (f(x)) \in K$ is a root of $f(x)$. Indeed, we have

$$f(\alpha) = f(x + (f(x))) = f(x) + (f(x)) = 0 + (f(x)) = 0. \quad (1.2)$$

■

Example 1.4. As a concrete example, let $f(x) = x^2 - 2 \in \mathbb{Q}[x]$. Then $f(x)$ is irreducible over $\mathbb{Q}$, and hence there exists a field extension $K/\mathbb{Q}$ such that $f(x)$ has a root in K . In fact, we can take $K = \mathbb{Q}[x]/(x^2 - 2)$. Let $\alpha = x + (x^2 - 2) \in K$. Then α is a root of $f(x)$. The reader may also verify that $\{1 + (x^2 - 2), \alpha\}$ is a basis of K over $\mathbb{Q}$.

In fact, if $f(x)$ is a polynomial of degree n , then the extension K/F constructed above has degree $[K : F] = n$, and a basis of K over F is given by $\{1, \alpha, \alpha^2, \dots, \alpha^{n-1}\}$, where $\alpha = x + (f(x)) \in K$. This is exactly the following theorem.

Theorem 1.5. *Let $p(x) \in F[x]$ be an irreducible polynomial of degree n , and let $K = F[x]/(p(x))$. Then $[K : F] = n$, and a basis of K over F is given by $\{1, \alpha, \alpha^2, \dots, \alpha^{n-1}\}$, where $\alpha = x + (p(x)) \in K$.*

Proof. Note that K , as a set, is precisely $\{a_0 + a_1\alpha + \dots + a_{n-1}\alpha^{n-1} \mid a_i \in F\}$, since any polynomial of degree n or higher can be reduced modulo $p(x)$ to a polynomial of degree less than n . Hence, the set $\{1, \alpha, \alpha^2, \dots, \alpha^{n-1}\}$ spans K over F . To see that this set is linearly independent, suppose we have a linear combination

$$a_0 + a_1\alpha + \dots + a_{n-1}\alpha^{n-1} = 0 \quad (1.3)$$

Since $p(x)$ is irreducible, the only way for this to happen is if all coefficients $a_i = 0$. Thus, the set $\{1, \alpha, \alpha^2, \dots, \alpha^{n-1}\}$ is linearly independent. ■

Example 1.6. Consider the polynomial $p(x) = x^2 + x + 1 \in \mathbb{F}_2[x]$. This polynomial is irreducible over $\mathbb{F}_2$. Let $K = \mathbb{F}_2[x]/(x^2 + x + 1) = \{a_0 + a_1\alpha \mid a_0, a_1 \in \mathbb{F}_2, \alpha^2 + \alpha + 1 = 0\}$. This K is an extension of degree 2 over $\mathbb{F}_2$,

Let K/F be an extension and let $\alpha \in K$. Denote by $F(\alpha)$ the smallest subfield of K containing both F and α . We say that $F(\alpha)$ is the subfield of K generated by α over F . Of course, we can have multiple generators, and we denote by $F(\alpha_1, \dots, \alpha_n)$ the smallest subfield of K containing F and $\alpha_1, \dots, \alpha_n \in K$. We say that $F(\alpha_1, \dots, \alpha_n)$ is the subfield of K generated by $\alpha_1, \dots, \alpha_n$ over F .

We say $F(\alpha_1, \alpha_2, \dots)$ is *finitely generated* if there exists a finite set of generators $\alpha_1, \dots, \alpha_n$ such that $F(\alpha_1, \dots, \alpha_n) = F(\alpha_1, \alpha_2, \dots)$. Otherwise, we say that $F(\alpha_1, \alpha_2, \dots)$ is *infinitely generated*. If the generating set has a single element α , we say that $F(\alpha)$ is a *simple extension* of F .

Theorem 1.7. *Let F be a field and $p(x)$ an irreducible polynomial over F . Suppose K/F is a field extension and K contains a root α of $p(x)$. Then $F(\alpha) \cong F[x]/(p(x))$.*

Proof. Define the map $\varphi : F[x] \rightarrow F(\alpha)$ by $f(x) \mapsto f(\alpha)$. This is a ring homomorphism. Clearly, $\varphi|_F = \text{id}_F$, and $(p(x)) \subseteq \ker \varphi$. Since $p(x)$ is irreducible, $(p(x))$ is a maximal ideal of $F[x]$, and hence $\ker \varphi = (p(x))$. This induces an injective map $\tilde{\varphi} : F[x]/(p(x)) \rightarrow F(\alpha)$ given by $f(x) + (p(x)) \mapsto f(\alpha)$. So $\text{Im } \tilde{\varphi}$ is a subfield of $F(\alpha)$ containing both α and F . By the very definition of $F(\alpha)$, we must have $\text{Im } \tilde{\varphi} = F(\alpha)$, showing φ is surjective. The isomorphism holds. ■

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